Universitat Pompeu Fabra Barcelona

MTG Music Technology Group

Music Technology Lab S103

Assignment 2: First model and inference strategies

Due: April the 19th at 11:59

This assignment focuses on developing a solid understanding of inference strategies, which will play a central role in the upcoming project. To begin, you are required to carefully study the Python script `FlowInference.py`, available at the SCAPES repository. You should aim to understand the structure and functionality of the code as thoroughly as possible. Note that the class and functions in such script are the ones we used when doing inference in the hands-on session.

Your report should first provide a clear description of the `FlowInference` class, explaining its purpose and how it operates within the overall system. You should then describe in detail the functions `load_and_encode`, `run_resynthesis_pipeline`, and `run_interpolation_pipeline`, highlighting their roles and how they contribute to the inference process. Particular attention should be given to the concept of the timeline, as it is a key element in the design and will be essential for your future work.

In parallel, you should reflect on your group's class experience with the model and identify the types of sounds it synthesizes most effectively. Based on this understanding, you will curate a dataset composed of at least 10 distinct types of sounds that are well-suited to the model's capabilities.

Using this curated dataset, you will train a SCAPES model following the notebooks introduced in class. This process includes applying the data preparation stage to your dataset, training the model, and experimenting with the different inference methods available. You are encouraged to use the large version of the model configuration whenever possible; however, if computational constraints arise, medium or small configurations are acceptable. In all cases, you should aim to train for at least 75 epochs.

Finally, you must document your training setup and results in detail. This includes describing your choices, reporting relevant metrics, and reflecting on the outcomes. Please follow the report template provided in the course folder to structure your submission.